



2009 Trial Examination

FORM VI

MATHEMATICS 2 UNIT

Tuesday 11th August 2009

General Instructions

- Reading time — 5 minutes
- Writing time — 3 Hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 120
- All ten questions may be attempted.
- All ten questions are of equal value.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the ten questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question 1.

6F: SJE
6Q: JMR

6G: FMW
6R: LYL

6H: BDD
6S: RCF

6P: KWM

Checklist

- SGS booklets — 10 per boy
- Candidature — 101 boys

Examiner

KWM

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

(a) Solve $2^x = \frac{1}{16}$.

1

(b) Factorise $x^3 + 27$.

1

(c) Simplify $\frac{x}{2} - \frac{x-1}{3}$.

2

(d) Find a primitive for $\sqrt{x}$.

1

(e) Evaluate $\sum_{k=1}^3 k^2$.

1

(f) Solve $\sin \alpha = -\frac{1}{2}$, for $0 \leq \alpha \leq 2\pi$.

2

(g) Solve $|x - 2| < 5$.

2

(h) Given $\frac{2}{\sqrt{5} + 2} = p\sqrt{5} - q$, find p and q .

2

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

- (a) Consider the series $32 + 36 + 40 + \dots + 92$.

(i) Show that the series is arithmetic.

1

(ii) How many terms are there in the series?

1

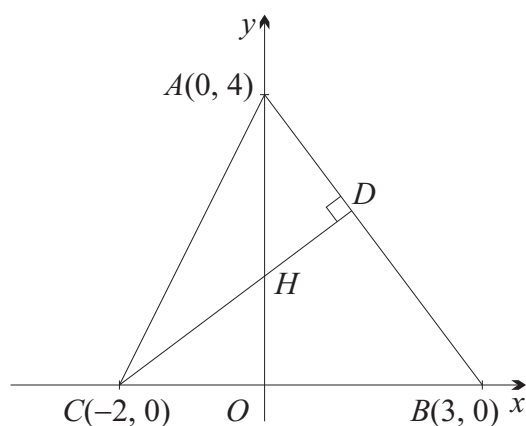
(iii) Find the sum of the series.

1

- (b) Gillian deposits \$12 000 in a fixed term investment account earning 6% p.a. compounded **monthly**. Calculate the value of her investment after five years. Give your answer correct to the nearest cent.

2

(c)



The triangle above has vertices $A(0, 4)$, $B(3, 0)$ and $C(-2, 0)$. AO and CD are the altitudes drawn from vertices A and C respectively.

(i) Find the gradient of the side AB .

1

(ii) Show that the side AB has equation $4x + 3y - 12 = 0$.

1

(iii) Calculate the perpendicular distance from the point $C(-2, 0)$ to the side AB .

2

(iv) Find the equation of the altitude CD .

2

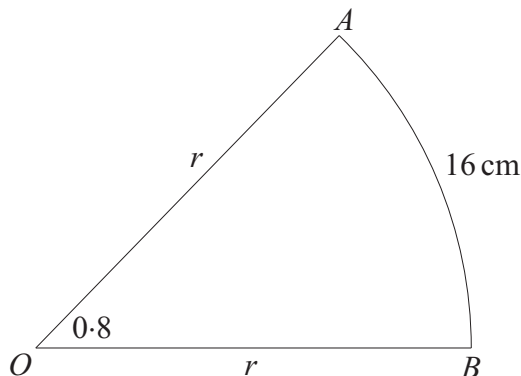
(v) Hence find the coordinates of the point H , the point of intersection of the altitudes AO and CD .

1

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

(a)



The diagram shows a sector of a circle. The arc AB is 16 cm, the radius is r cm and $\angle AOB = 0.8$ radians.

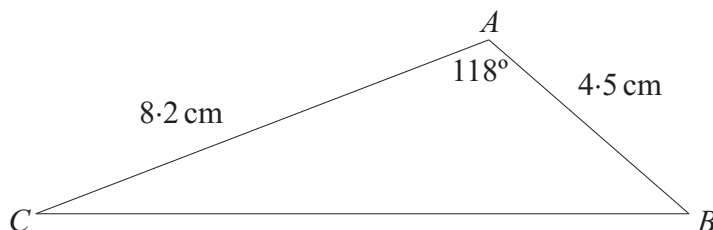
(i) Find the value of r .

1

(ii) Calculate the area of the sector.

2

(b)



The diagram above shows $\triangle ABC$ where $AB = 4.5$ cm, $AC = 8.2$ cm and $\angle CAB = 118^\circ$.

(i) Find the length of side BC , correct to the nearest millimetre.

2

(ii) Calculate the area of $\triangle ABC$ in cm^2 , correct to one decimal place.

1

(c) Differentiate the following functions:

(i) $y = \frac{1}{x}$

1

(ii) $y = \tan 2x$

1

(iii) $y = xe^x$

2

(iv) $y = \frac{\log_e x}{x}$

2

Exam continues next page ...

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

(a) Simplify $2\log_3 6 - \log_3 4$.

2

(b) Find:

(i) $\int_0^{\ln 3} e^x dx$

2

(ii) $\int_0^1 \frac{x}{x^2 + 1} dx$

2

(c) Find the equation of the tangent to the curve $y = \cos(\pi - x)$ at the point where $x = \frac{\pi}{3}$.

3

(d) Consider the parabola $x^2 - 2x + 4y + 9 = 0$.

(i) Express the equation in the form $(x - h)^2 = -4a(y - k)$.

1

(ii) Find the coordinates of the focus.

1

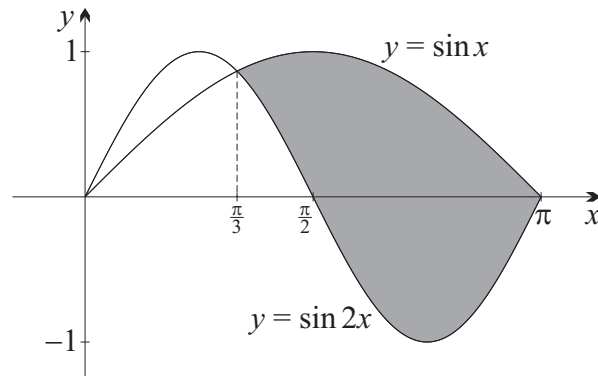
(iii) Write down the equation of the directrix.

1

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

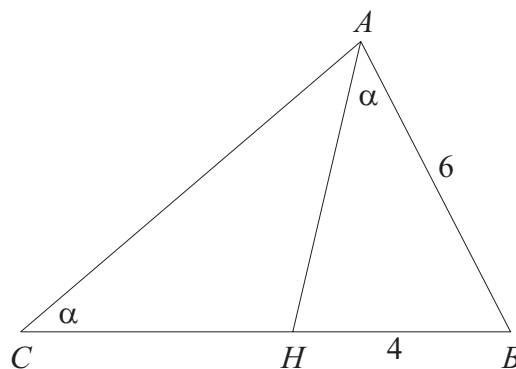
(a)



The diagram above shows the curves $y = \sin 2x$ and $y = \sin x$ for $0 \leq x \leq \pi$, intersecting at $x = 0$, $x = \frac{\pi}{3}$ and $x = \pi$. Find the exact area of the shaded region bounded by the two curves.

3

(b)



In the diagram above, $\angle BCA = \angle BAH = \alpha$, $AB = 6$ and $BH = 4$.

(i) Show that $\triangle ABC \sim \triangle HBA$.

2

(ii) Hence, or otherwise, find the length HC .

2

(c) Consider the quadratic equation $x^2 - 2kx + (8k - 15) = 0$.

(i) Find the discriminant and write it in simplest form.

1

(ii) For what values of k does the equation have real roots?

2

(iii) If three times the sum of the roots is equal to twice the product of the roots, find the value of k .

2

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**(a) Consider the function $h(x) = \sqrt{x^2 - 1}$.(i) Show that $h(x)$ is an even function.**1**(ii) Find the domain of $h(x)$.**1**(b) Consider the function $y = (x + 1)^3(x - 3)$.(i) Use the product rule to show that $\frac{dy}{dx} = 4(x + 1)^2(x - 2)$.**2**

(ii) Find the coordinates of the stationary points and determine their nature.

4

(iii) The curve has a point of inflexion where the tangent is not horizontal. Find the coordinates of this point.

2(iv) Sketch the curve $y = (x + 1)^3(x - 3)$, showing all the important features.**2****QUESTION SEVEN** (12 marks) Use a separate writing booklet.**Marks**(a) The population P of mosquitoes in a laundry is growing exponentially according to the equation $P = 50e^{kt}$, where t is the time in days after the insects are first counted. After four days the population has doubled.(i) Find the exact value of the constant k .**2**

(ii) How many mosquitoes will there be after 10 days?

2

(iii) At what rate is the population increasing after 10 days?

1(b) (i) Copy and complete the table correct to four decimal places where necessary for the function $y = \log_e(x + 1)$.**1**

x	0	0.5	1	1.5	2
y					

(ii) Use Simpson's rule with 5 function values to find an approximation to

2 $\int_0^2 \log_e(x + 1) dx$. Write your answer correct to three decimal places.(iii) Show that $\frac{d}{dx}((x + 1) \log_e(x + 1) - x) = \log_e(x + 1)$.**2**(iv) Hence find the exact value of $\int_0^2 \log_e(x + 1) dx$, and determine whether or not your approximation in part (ii) is accurate to three decimal places.**2****Exam continues overleaf ...**

QUESTION EIGHT (12 marks) Use a separate writing booklet.**Marks**

- (a) A swimming pool is being emptied. The volume of water L litres in the pool after t minutes is given by the equation

$$L = 1000(20 - t)^3.$$

- (i) Find the rate at which the pool is emptying after 10 minutes. 2

- (ii) When is the pool emptying at a maximum rate? 1

- (b) (i) Expand $(\sqrt{3}u - 1)(u - \sqrt{3})$. 1

- (ii) Hence solve $\sqrt{3}\tan^2 \theta - 4\tan \theta + \sqrt{3} = 0$, for $0 \leq \theta \leq 2\pi$. 2

- (c) A particle moves in a straight line so that after t seconds ($t \geq 0$) its velocity v is given by $v = \left(\frac{2}{1+t} - t\right)$ m/s. The displacement of the particle from the origin is given by x metres.

- (i) Find the acceleration of the particle when $t = 0$. 2

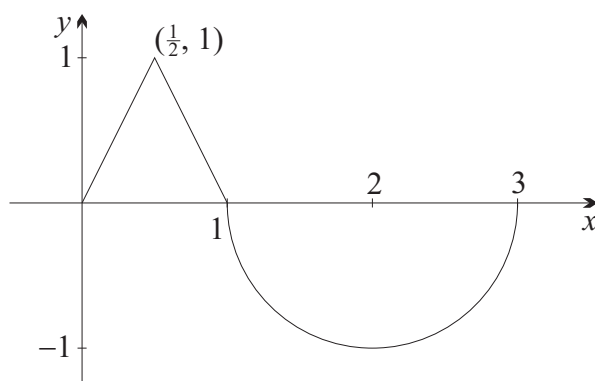
- (ii) If the particle is initially at the origin, find the displacement as a function of t . 2

- (iii) When is the particle stationary? 1

- (iv) How far does the particle travel in the first 2 seconds? Give your answer correct to three significant figures. 1

QUESTION NINE (12 marks) Use a separate writing booklet.**Marks**

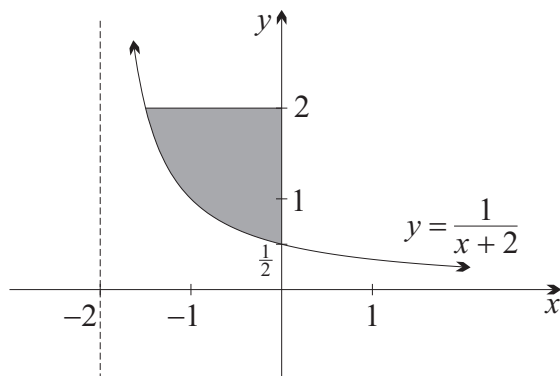
- (a)



The diagram above shows the function $y = g(x)$ with domain $0 \leq x \leq 3$. The arc is a semi-circle. Find $\int_0^3 g(x) dx$. 1

QUESTION NINE (Continued)

(b)



The diagram above shows the curve $y = \frac{1}{x+2}$ for $x > -2$.

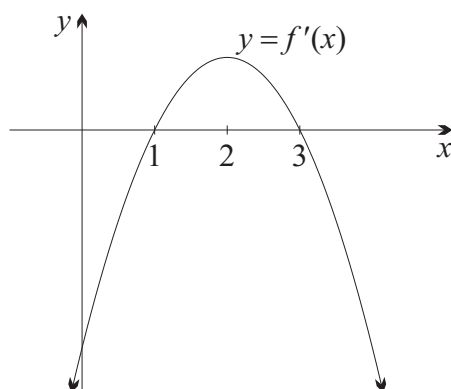
(i) Show that $x^2 = \frac{1}{y^2} - \frac{4}{y} + 4$.

2

(ii) Calculate the exact volume of the solid of revolution formed when the shaded region bounded by the y -axis, the line $y = 2$ and the curve is rotated about the y -axis.

4

(c)



The diagram above shows the graph of the gradient function $y = f'(x)$ of the function $y = f(x)$.

(i) For which values of x is the curve $y = f(x)$ increasing?

1

(ii) For which values of x is the curve $y = f(x)$ concave up?

1

(iii) Given $f(0) = f(2) = f(4) = 0$, sketch the curve $y = f(x)$ for $0 \leq x \leq 4$.

2

QUESTION TEN (12 marks) Use a separate writing booklet.**Marks**

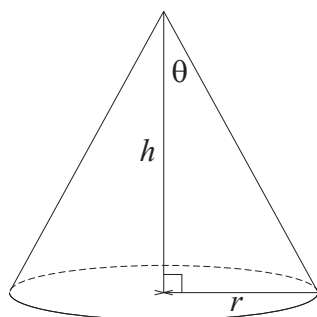
- (a) Katherine borrows \$200 000 from the bank. The loan plus the interest is to be repaid in equal monthly instalments of \$ M over 25 years. Reducible interest is charged at 6% p.a. and is calculated monthly. Let A_n be the amount owing after n months.

(i) Write down expressions for A_1 and A_2 , and show that the amount owing after 3 months is given by $A_3 = 200\,000(1.005)^3 - M(1 + 1.005 + 1.005^2)$. **3**

(ii) Hence write an expression for A_n . **1**

(iii) Calculate the monthly instalment \$ M correct to the nearest cent. **2**

(b)



A right circular cone of radius r and height h has a total surface area S and volume V . Note that $S = \pi r^2 + \pi r\sqrt{r^2 + h^2}$ and $V = \frac{1}{3}\pi r^2 h$.

(i) Show that $9V^2 = r^2(S^2 - 2\pi r^2 S)$. **2**

(ii) For a fixed surface area S , find $\frac{d}{dr}(9V^2)$. **1**

(iii) Hence find the semi-vertical angle θ that gives the maximum volume of the cone for a fixed surface area S . Write your answer correct to the nearest minute. **3**

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION 1

$$(a) \quad 2^x = \frac{1}{16} \\ x = -4 \quad \checkmark$$

$$(b) \quad x^3 + 27 = (x+3)(x^2 - 3x + 9) \quad \checkmark$$

$$(c) \quad \frac{x}{2} - \left(\frac{x-1}{3}\right) = \frac{3x - 2(x-1)}{6} \quad \checkmark \\ = \frac{3x - 2x + 2}{6} \\ = \frac{x+2}{6} \quad \checkmark$$

$$(d) \quad f(x) = x^{1/2} \\ F(x) = \frac{2}{3} x^{3/2} + C \quad \checkmark$$

$$(e) \quad \sum_{k=1}^3 k^2 = 1^2 + 2^2 + 3^2 \\ = 14 \quad \checkmark$$

$$(f) \quad \sin \alpha = -\frac{1}{2} \quad \frac{\swarrow}{\vee} \frac{\swarrow}{\vee}$$

$$\alpha = \pi + \frac{\pi}{6} \quad \text{or} \quad \alpha = 2\pi - \frac{\pi}{6}$$

$$\alpha = \frac{7\pi}{6} \quad \checkmark \quad \text{or} \quad \alpha = \frac{11\pi}{6} \quad \checkmark$$

$$(g) \quad |x-2| < 5$$

$$x-2 < 5 \quad \text{or} \quad x-2 > -5$$

$$x < 7 \quad \checkmark \quad \text{or} \quad x > -3 \quad \checkmark$$

$$-3 < x < 7$$

(h)

$$\frac{2}{\sqrt{5}+2} = p\sqrt{5} - q$$

$$\frac{2}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} = p\sqrt{5} - q$$

$$\frac{2\sqrt{5}-4}{5-4} = p\sqrt{5} - q$$

$$2\sqrt{5}-4 = p\sqrt{5} - q$$

Hence $p=2$ and $q=4 \quad \checkmark$

(12)

QUESTION 2

(a) $32 + 36 + 40 + \dots + 92$

(i) $36 - 32 = 40 - 36 = 4$

$\therefore$ an AP. ✓

($a = 32$ and $d = 4$) ✓

(ii) $a + (n-1)d = 92$

$32 + (n-1)4 = 92$

$4n + 28 = 92$

$4n = 64$

$n = 16$ ✓

There are 16 terms.

(iii) $S_n = \frac{n}{2}(a+l)$

$S_{16} = \frac{16}{2}(32 + 92)$

$S_{16} = 992$ ✓

(b) $R = 1.005$

$P = 12\,000$

$n = 60$

$A = PR^n$

$= 12\,000 (1.005)^{60}$ ✓

$= \text{R}16,186.20$ ✓

(c) $A(0,4)$, $B(3,0)$

(i) gradient $AB = -\frac{4}{3}$ ✓

(ii) $(0,4)$, $m = -\frac{4}{3}$

$y - y_1 = m(x - x_1)$

$y - 4 = -\frac{4}{3}(x - 0)$ ✓

$3y - 12 = -4x$

$4x + 3y - 12 = 0$

(as required).

(iii) $4x + 3y - 12 = 0$

$C(-2,0)$

$d = \frac{|-8 + 0 - 12|}{\sqrt{16 + 9}}$ ✓

$d = \frac{20}{5}$

$d = 4$ units. ✓

(iv) gradient $CD = \frac{3}{4}$ ✓ $C(-2,0)$

$y - y_1 = m(x - x_1)$

$y = \frac{3}{4}(x + 2)$

$4y = 3x + 6$

$3x - 4y + 6 = 0$ ✓

(v) put $x = 0$.

$3x - 4y + 6 = 0$

$-4y + 6 = 0$

$4y = 6$

$y = \frac{3}{2}$

$H(0, \frac{3}{2})$ // ✓

12

QUESTION 3

(a) (i) $s = r\theta$
 $16 = 0.8 \times r$
 $r = 20 \text{ cm} \checkmark$

(ii) $A = \frac{1}{2} r^2 \theta \checkmark$
 $= \frac{1}{2} \times 400 \times 0.8$
 $= 160 \text{ cm}^2 \checkmark$

(b) (i) $BC^2 = 8.2^2 + 4.5^2 - 2 \times 8.2 \times 4.5 \cos 118^\circ \checkmark$
 $BC \doteq 11.1 \text{ cm} \checkmark$

(ii) $\text{Area} = \frac{1}{2} ab \sin C$
 $= \frac{1}{2} \times 8.2 \times 4.5 \sin 118^\circ$
 $\doteq 16.3 \text{ cm}^2 \checkmark$

(c) (i) $y = x^{-1}$
 $\frac{dy}{dx} = -\frac{1}{x^2} \checkmark$

(ii) $y = \tan 2x$
 $\frac{dy}{dx} = 2 \sec^2 2x \checkmark$

(iii) $y = xe^x$
 $\frac{dy}{dx} = e^x + xe^x \checkmark \checkmark$
 $= e^x(1+x)$

(iv) $y = \frac{\ln x}{x}$
 $\frac{dy}{dx} = \frac{x \times \frac{1}{x} - \ln x \times 1}{x^2} \checkmark \checkmark$

$$\frac{dy}{dx} = \frac{1 - \ln x}{x^2}$$

(12)

QUESTION 4

(a) $2\log_3 6 - \log_3 4$
 $= \log_3 36 - \log_3 4 \checkmark$
 $= \log_3 9$
 $= 2 \checkmark$

(b)
 (i) $\int_0^{\ln 3} e^x dx = \left[e^x \right]_0^{\ln 3} \checkmark$
 $= e^{\ln 3} - e^0$
 $= 3 - 1$
 $= 2 \checkmark$

(ii) $\int_0^1 \frac{x}{x^2+1} dx = \frac{1}{2} \left[\ln(x^2+1) \right]_0^1 \checkmark$
 $= \frac{1}{2} \left\{ \ln 2 - \ln 1 \right\}$
 $= \frac{1}{2} \ln 2 \checkmark$
 $= \ln \sqrt{2}.$

(c) $y = \cos(\pi - x)$
 $\left(\frac{\pi}{3}, -\frac{1}{2} \right) \checkmark$
 $\frac{dy}{dx} = \sin(\pi - x) \checkmark$
 at $x = \frac{\pi}{3}$, gradient $= \sin \frac{2\pi}{3}$
 $= \frac{\sqrt{3}}{2}$
 $y - y_1 = m(x - x_1)$
 $y + \frac{1}{2} = \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{3} \right) \checkmark$

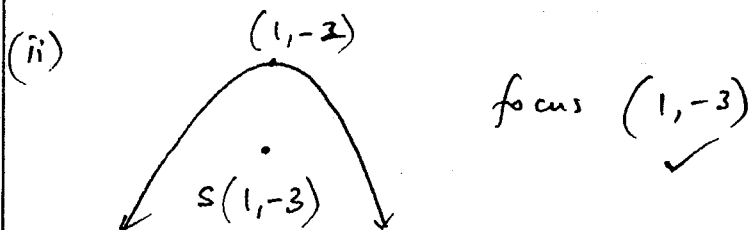
$$2y + 1 = \sqrt{3}x - \frac{\sqrt{3}\pi}{3}$$

$$\sqrt{3}x - 2y - 1 - \frac{\sqrt{3}\pi}{3} = 0$$

(d) $x^2 - 2x + 4y + 9 = 0$

(i) $(x-1)^2 - 1 + 4y + 9 = 0$
 $(x-1)^2 = -4y - 8$
 $(x-1)^2 = -4(y+2) \checkmark$

vertex $(1, -2)$, $a = 1.$



(iii) $y = -1. \checkmark$

12

QUESTION 5

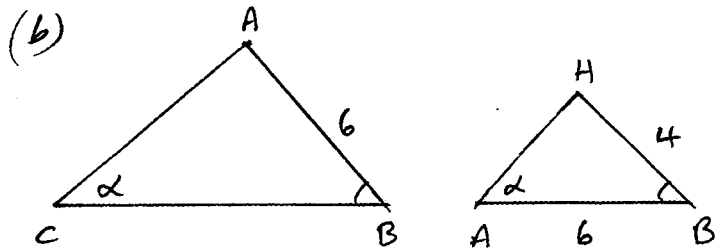
(a)
$$A = \int_{\frac{\pi}{3}}^{\pi} \sin x - \sin 2x \, dx$$

$$= \left[-\cos x + \frac{1}{2} \cos 2x \right]_{\frac{\pi}{3}}^{\pi}$$

$$= \left(1 + \frac{1}{2} \right) - \left(-\frac{1}{2} + \frac{1}{2} \times -\frac{1}{2} \right)$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4}$$

$A = 2\frac{1}{4}$ sq. units. ✓



(i) $\angle ACB = \angle HBA = \alpha$ (given) ✓
 $\angle ABC = \angle HBA$ (common) ✓
 $\therefore \triangle ABC \parallel \triangle HBA$ (A.A) ✓

(ii) $\frac{BC}{6} = \frac{6}{4}$ ✓ (matching sides in the ratio.)

$$BC = \frac{36}{4}$$

$$BC = 9$$

$$HC = BC - 4$$

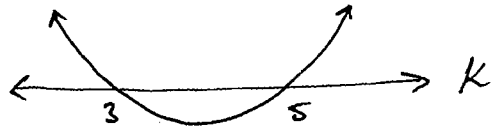
$$= 9 - 4$$

$HC = 5$ units. ✓

(c) $x^2 - 2Kx + (8K - 15) = 0$

(i) $\Delta = b^2 - 4ac$
 $\Delta = 4K^2 - 4(8K - 15)$
 $\Delta = 4K^2 - 32K + 60$ ✓

(ii) $4K^2 - 32K + 60 \geq 0$
 $K^2 - 8K + 15 \geq 0$
 $(K - 3)(K - 5) \geq 0$ ✓



$K \leq 3, K \geq 5$ ✓

(iii) $\alpha + \beta = -\frac{b}{a}$
 $= \frac{2K}{1}$

$\alpha\beta = \frac{c}{a}$
 $= 8K - 15$ ✓

$6K = 2(8K - 15)$
 $6K = 16K - 30$
 $-10K = -30$
 $K = 3$ ✓

QUESTION 6

(a) (i) $h(x) = \sqrt{x^2 - 1}$
 $h(-x) = \sqrt{(-x)^2 - 1}$
 $= \sqrt{x^2 - 1}$
 $= h(x) \quad \checkmark$
 $\therefore h(x)$ is even.

(ii) Domain $x^2 - 1 \geq 0$
 $(x-1)(x+1) \geq 0$
 $x \leq -1, x \geq 1 \quad \checkmark$

(b) $y = (x+1)^3(x-3)$

i) $y' = 3(x+1)^2(x-3) + (x+1)^3 \checkmark$
 $= (x+1)^2 \{ 3x - 9 + x + 1 \} \checkmark$
 $= (x+1)^2 (4x - 8)$
 $= 4(x+1)^2(x-2)$
 (as required.)

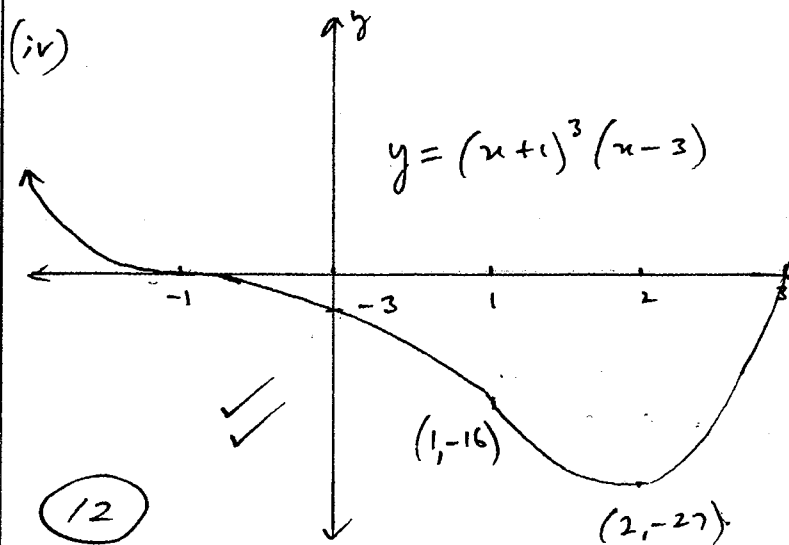
(ii) $4(x+1)^2(x-2) = 0$
 $x = -1$ or $x = 2$
 stationary points.
 $(-1, 0) \checkmark$ and $(2, -27) \checkmark$
 - nature.

x	-2	-1	0	0	2	3
y'	-16	0	-8	-8	0	64

$(-1, 0)$ is a horizontal pt. of inflex. $\checkmark$

$(2, -27)$ is a minimum stationary point. $\checkmark$

(iii) $y'' = 8(x+1)(x-2) + 4(x+1)^2$
 $= 4(x+1) \{ 2(x-2) + (x+1) \} \checkmark$
 $= 4(x+1)(3x-3)$
 $12(x+1)(x-1) = 0$
 $x = -1$ or $x = 1$
 $(-1, 0)$ and $(1, -16) \checkmark$ is the other pt. of inflexion.



QUESTION 7

(a) $P = 50e^{Kt}$
 (i) when $t=4$, $P=100$.
 $100 = 50e^{4K}$ ✓
 $2 = e^{4K}$
 $4K = \ln 2$
 $K = \frac{1}{4} \ln 2$ ✓

(ii) put $t=10$.
 $P = 50e^{\frac{10}{4} \ln 2}$
 $P = 50e^{\frac{5}{2} \ln 2}$ ✓
 $P = 50(e^{\ln 2})^{\frac{5}{2}}$
 $P = 50 \times 2^{\frac{5}{2}}$
 $P = 282.8$ (20052)
 approximately 283 ✓
 mosquitoes.

(iii) $\frac{dP}{dt} = KP$
 $\frac{dP}{dt} = \frac{1}{4} \ln 2 \times 282.8$
 $\frac{dP}{dt} \doteq 49$ mosquitoes/day
 ✓ (5052 ln 2)

(b) (i)

x	0	0.5	1	1.5	2
y	0	0.4055	0.6931	0.9163	1.0986

$f(x) = \ln(x+1)$

i) $h = \frac{1}{2}$.

$\int_0^2 \ln(x+1) dx \doteq$

$\frac{1}{6} \{ 0 + 4 \times 0.4055 + 0.6931 \}$ ✓

$+ \frac{1}{6} \{ 0.6931 + 4 \times 0.9163 + 1.0986 \}$

$= 1.295$ ✓ (3 dec. places)

(iii)

$\frac{d}{dx} \left((x+1) \ln(x+1) - x \right)$
 $= 1 \times \ln(x+1) + (x+1) \times \frac{1}{(x+1)} - 1$ ✓
 $= \ln(x+1) + 1 - 1$
 $= \ln(x+1)$ (as required).

(iv) $\int_0^2 \ln(x+1) dx = \left[(x+1) \ln(x+1) - x \right]_0^2$
 $= (3 \ln 3 - 2) - (\ln 1 - 0)$
 $= 3 \ln 3 - 2$ ✓
 $\doteq 1.296$ (3 decimal places)

The approximation is not ✓
 accurate to 3 decimal places.

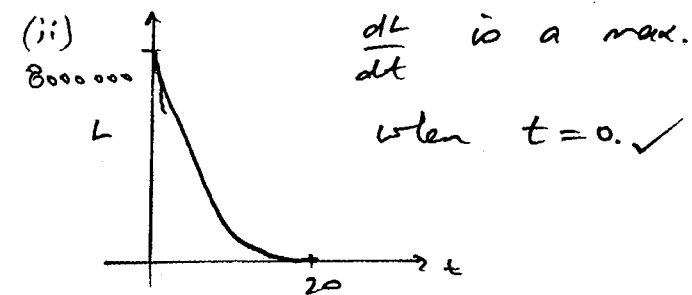
(12)

QUESTION 8

(a) $L = 1000(20-t)^3$
 (i) $\frac{dL}{dt} = -3000(20-t)^2 \checkmark$

at $t=10$, $\frac{dL}{dt} = -300000$

After 10 minutes the pool is being emptied at 300 000 L/minute. $\checkmark$



The pool is being emptied at a maximum rate initially.

(b) (i) $(\sqrt{3}u - 1)(u - \sqrt{3})$
 $= \sqrt{3}u^2 - 3u - u + \sqrt{3} \checkmark$
 $= \sqrt{3}u^2 - 4u + \sqrt{3}$

(ii) $\sqrt{3}\tan^2\theta - 4\tan\theta + \sqrt{3} = 0$
 $(\sqrt{3}\tan\theta - 1)(\tan\theta - \sqrt{3}) = 0$
 $\tan\theta = \frac{1}{\sqrt{3}} \quad \text{or} \quad \sqrt{\tan\theta} = \sqrt{3}$
 $\theta = \frac{\pi}{6}, \frac{7\pi}{6} \quad \text{or} \quad \theta = \frac{\pi}{3}, \frac{4\pi}{3} \checkmark$

(c)

$v = \frac{2}{1+t} - t \quad \text{m/s}$

(i) $\frac{dv}{dt} = \frac{-2}{(1+t)^2} - 1$

$a = \frac{-2}{(1+t)^2} - 1 \checkmark$

when $t=0$.

$a = -2 - 1$

$a = -3 \text{ m/s}^2 \checkmark$

(ii) $\frac{dx}{dt} = \frac{2}{1+t} - t$

$x = 2\ln(1+t) - \frac{t^2}{2} + c \checkmark$

when $t=0$, $x=0$, $\therefore c=0$

$x = 2\ln(1+t) - \frac{t^2}{2} \checkmark$

(iii) $v=0$. $\frac{2}{1+t} - t = 0$

$2 - t(1+t) = 0$

$2 - t - t^2 = 0$

$t^2 + t - 2 = 0$

$(t-1)(t+2) = 0$

$t=1$ or $t=-2$ (omit) $\checkmark$

The particle is stationary at $t=1$ s

(iv)

x	0	0.8863	0.1972
t	0	1	2

$x = 2\ln(1+t) - \frac{t^2}{2}$

distance travelled =

$0.8863 + (0.8863 - 0.1972)$

$= 1.5754$

$= 1.58 \text{ m (3 sig. fig.)} \checkmark$

12

QUESTION 9

(a) $\int_0^3 g(x) dx = \frac{1}{2}bh - \frac{1}{2}\pi r^2 \checkmark$
 $= \frac{1}{2} - \frac{1}{2}\pi$
 $= \frac{1}{2}(1 - \pi) \checkmark$

(b)

(i) $y = \frac{1}{x+2}$
 $x+2 = \frac{1}{y}$
 $x = \frac{1}{y} - 2 \checkmark$
 $x^2 = \frac{1}{y^2} - \frac{4}{y} + 4 \checkmark$

(ii)

$V = \pi \int_{\frac{1}{2}}^2 \left(\frac{1}{y^2} - \frac{4}{y} + 4 \right) dy$
 $= \pi \left[-\frac{1}{y} - 4 \ln y + 4y \right]_{\frac{1}{2}}^2 \checkmark$

$= \pi \left\{ \left(-\frac{1}{2} - 4 \ln 2 + 8 \right) - \left(-2 - 4 \ln \frac{1}{2} + 2 \right) \right\}$

$= \pi \left\{ -\frac{1}{2} - 4 \ln 2 + 8 + 2 - 4 \ln 2 - 2 \right\} \checkmark$

$= \pi (7\frac{1}{2} - 8 \ln 2) \checkmark$

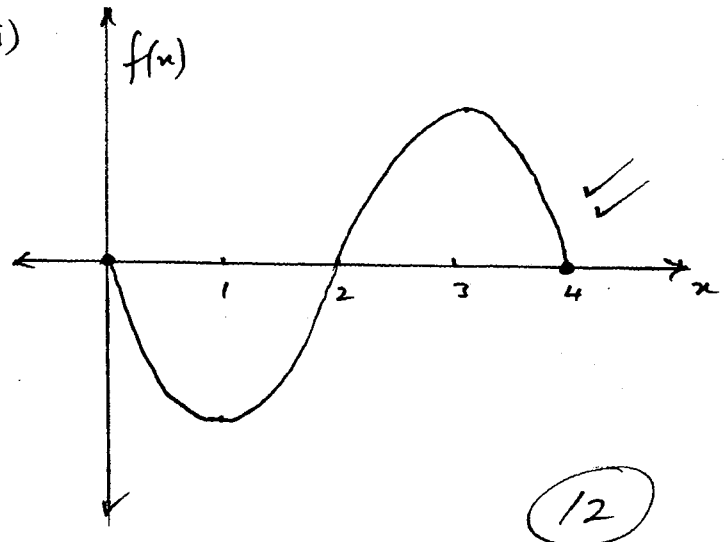
$= \frac{\pi}{2} (15 - 16 \ln 2) \text{ cubic units}$

(c)

(i) $1 < x < 3 \checkmark$

(ii) $x < 2 \checkmark$

(iii)



QUESTION 10

(a) $R = 1.005$
 (i) $A_1 = 200\,000(1.005) - M$ ✓

$$A_2 = (200\,000(1.005) - M)1.005 - M$$

$$A_2 = 200\,000(1.005)^2 - M(1.005) - M$$
 ✓

$$A_3 = A_2 \times 1.005 - M$$

$$= 200\,000(1.005)^3 - M(1.005)^2 - M(1.005) - M$$
 ✓

$$\frac{A_n}{3} = 200\,000(1.005)^3 - M\{1 + (1.005) + (1.005)^2\}$$

(as required).

(ii)

$$A_n = 200\,000(1.005)^n - M\{1 + (1.005) + (1.005)^2 + \dots + (1.005)^{n-1}\}$$

Using $S_n = a \frac{(r^n - 1)}{r - 1}$,

$$A_n = 200\,000(1.005)^n - \frac{M(1.005^n - 1)}{0.005}$$
 ✓

Put $A_n = 0$ and $n = 300$.

$$M = \frac{200\,000(1.005)(0.005)}{(1.005)^{300} - 1}$$
 ✓

$$M = \$1\,288.60$$
 ✓

(b)(i) $V = \frac{1}{3}\pi r^2 h$
 $3V = \pi r^2 h$
 $9V^2 = \pi^2 r^4 h^2$

now $S = \pi r^2 + \pi r(r^2 + h^2)^{1/2}$

$$S - \pi r^2 = \pi r(r^2 + h^2)^{1/2}$$
 ✓

square both sides.

$$S^2 - 2S\pi r^2 + \pi^2 r^4 = \pi^2 r^2 (r^2 + h^2)$$

$$S^2 - 2S\pi r^2 + \pi^2 r^4 = \pi^2 r^4 + \pi^2 r^2 h^2$$

$$S^2 - 2S\pi r^2 = \pi^2 r^2 h^2$$

$$S^2 - 2S\pi r^2 = \frac{\pi^2 r^4 h^2}{r^2}$$
 ✓

$$S^2 - 2S\pi r^2 = \frac{9V^2}{r^2}$$

$\therefore 9V^2 = r^2(S^2 - 2S\pi r^2)$
 as required.

(ii)

$$9V^2 = r^2 S^2 - 2S\pi r^4$$

$$\frac{d}{dr}(9V^2) = 2rS^2 - 8S\pi r^3$$
 ✓

(iii) The maximum value of $9V^2$ will give the radius required for the maximum value of V .

$$2rS^2 - 8S\pi r^3 = 0$$

$$2rS(S - 4\pi r^2) = 0$$

$$S = 0 \quad \text{or} \quad S = 4\pi r^2$$
 ✓

(omit)

Now $S = \pi r^2 + \pi r(r^2 + h^2)^{1/2}$

$$4\pi r^2 = \pi r^2 + \pi r(r^2 + h^2)^{1/2}$$

$$3\pi r^2 = \pi r(r^2 + h^2)^{1/2}$$

$$9\pi^2 r^4 = \pi^2 r^2 (r^2 + h^2)$$

$$9\pi^2 r^4 = \pi^2 r^4 + \pi^2 r^2 h^2$$

$$8r^2 = r^2 + h^2$$
 ✓

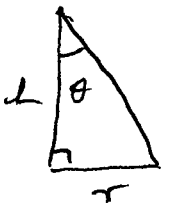
$$8r^2 = h^2$$

$$\frac{r^2}{h^2} = \frac{1}{8}$$

$$\frac{r}{h} = \frac{1}{2\sqrt{2}}$$

$$\tan \theta = \frac{1}{2\sqrt{2}}$$

$$\theta = 19^\circ 28'$$
 ✓



(12)

$$\frac{d}{dr^2}(9V^2) = 2S^2 - 24\pi r^2 S$$

(when $S = 4\pi r^2$)

$$= -64\pi r^2 < 0$$

$\therefore$ a max. exists at $S = 4\pi r^2$